

SIMCAST-Spatial — Project Overview

Piero Palacios

Henry Juarez

SIMCAST-Spatial — Project Overview (for non-coders)

What is it?

SIMCAST-Spatial is a scientific data-processing pipeline that turns climate data (temperature, humidity proxy, rainfall) into **spatial decision-support outputs** related to **potato late blight risk** and **fungicide application timing**.

It does this by: - Converting gridded climate datasets into values at many target locations (a point grid). - Reconstructing **hourly** temperature and relative humidity conditions from daily climate variables. - Running the **SIMCAST** disease-risk model at each location and time period. - Producing tabular summaries and map-ready raster outputs (GeoTIFFs), plus optional animations.

Even though it is implemented in Python, you can think of it as a “workflow” that: 1) prepares climate inputs, 2) computes disease pressure metrics, and 3) writes outputs you can map and interpret.

What is it for?

This project is designed to support **agroclimatic zoning and risk mapping** for potato late blight by answering questions like:

- Where (spatially) and when (seasonally) do weather conditions favor late blight development?
- How many fungicide applications would be expected under SIMCAST rules?
- Where are trends increasing/decreasing over the years?

Typical use cases: - **Regional risk mapping** for agricultural planning. - **Comparing seasons or regions** (e.g., wet vs dry seasons, highlands vs lowlands). - **Monitoring interannual variability** and **trend detection** over multiple years.

What inputs does it need (conceptually)?

You provide two main types of inputs:

1) A spatial grid of locations

A file containing many geographic features (usually points), each with an identifier (ID). Each point represents a location where the model is evaluated.

2) Climate time series on a grid (NetCDF)

Monthly files with daily values for: - Daily minimum temperature (**TMIN**) - Daily maximum temperature (**TMAX**) - Daily dew point temperature (**TDEW**) — used to estimate humidity - Daily precipitation (**PP**)

These are standard climate variables often available from reanalysis products, downscaled climate grids, or climate services.

The science and math behind the pipeline (plain language)

Step A — Align climate data to a common grid (resampling)

Climate datasets often come on different grids/resolutions. Before you can compare or combine them at the same location, they must be aligned.

Resampling means interpolating/transforming one gridded dataset so its pixels match a target grid: - Same pixel size (resolution) - Same pixel alignment - Same geographic coverage/projection (as needed)

In this pipeline, resampling is especially used to ensure precipitation matches the grid used for the temperature products.

Why this matters: if variables are misaligned, you could combine a temperature value from one pixel with rainfall from a different pixel—introducing spatial errors.

Step B — Extract climate values at each location (masking)

Once rasters are aligned, the pipeline “samples” the climate grids at each location in your point grid.

The result is a time series per location: - For each day: TMIN, TMAX, TDEW, and PP

These extracted values are stored efficiently in columnar tables (Parquet files) for fast reading later.

Step C — Reconstruct hourly conditions from daily climate data

SIMCAST relies on **hours of high relative humidity**, but many climate datasets are daily.

To bridge that gap, the pipeline reconstructs an **hourly** temperature and humidity profile using two standard components: 1) a **diurnal temperature curve model** (García-1), and 2) a **humidity-from-dew-point formulation** (Buck equation for vapor pressure).

1) A diurnal (day–night) temperature curve (García-1)

Given daily **TMIN** and **TMAX**, the pipeline approximates the 24-hour temperature pattern across the day using a smooth, piecewise curve that depends on: - Sunrise time H_n and sunset time H_o (computed from latitude/longitude and date) - Daily temperature extrema $T_{\min}$, $T_{\max}$ - The next day’s minimum temperature (to shape the overnight cooling)

Output: an estimated **hourly temperature** series $T(h)$ for $h = 1, 2, \dots, 24$.

2) Relative humidity from temperature + dew point (Buck equation)

Dew point is a strong indicator of atmospheric moisture. The pipeline computes: - Saturation vapor pressure at air temperature - Actual vapor pressure at dew point - Relative humidity as a ratio of the two

This is standard meteorological math used in many models.

Output: an estimated **hourly relative humidity** series $RH(h)$ for $h = 1, 2, \dots, 24$.

3) Hours of high humidity (HHR)

For each day, the model counts how many hours exceed a humidity threshold (default **90% RH**). This is often called **Hours of High Relative Humidity**.

Important detail: the project uses a 24-hour “observation window” from **13:00 today to 12:00 next day** (a convention inherited from the original workflow). This matters because humidity often peaks overnight; the window captures that pattern in a consistent way.

Step C — model details (math)

This subsection names the models explicitly and shows the equations used in the pipeline.

C.1 García-1 diurnal temperature reconstruction (conceptual)

The goal is to reconstruct hourly temperature $T(h)$ from daily values $T_{\min}, T_{\max}$ and solar timing H_n (sunrise), H_o (sunset). García-1 is implemented as a **piecewise** curve that smoothly models: - daytime warming after sunrise, - a late-afternoon transition, - nighttime cooling after sunset toward the next minimum.

(For non-coders: the key point is that the project does **not** assume temperature is constant within a day; it builds a realistic hourly cycle.)

C.2 Buck equation (vapor pressure) and relative humidity

Let $T(h)$ be the reconstructed hourly air temperature (°C), and let T_{dew} be dew point temperature (°C). Using the Buck-type saturation vapor pressure formulation (in kPa):

$$e_s(T) = 0.61121 \cdot \exp \left(\left(18.678 - \frac{T}{234.5} \right) \cdot \frac{T}{257.14 + T} \right)$$

Actual vapor pressure is computed from dew point:

$$e(T_{dew}) = 0.61121 \cdot \exp \left(\left(18.678 - \frac{T_{dew}}{234.5} \right) \cdot \frac{T_{dew}}{257.14 + T_{dew}} \right)$$

Relative humidity per hour is then:

$$RH(h) = 100 \cdot \frac{e(T_{dew})}{e_s(T(h))}$$

C.3 Hours of High Relative Humidity (HHR)

Given a threshold RH_{thr} (default 90%), the daily HHR metric is:

$$HHR = \sum_{h=1}^{24} \mathbb{1} [RH(h) \geq RH_{thr}]$$

This *HHR* (and the temperature during those high-humidity hours) is what feeds the SIMCAST blight-risk scoring.

Step D — Compute disease pressure and spray triggers (SIMCAST core)

SIMCAST is a rule-based model that turns weather conditions into:

BU (Blight Units)

A daily score based on: - **How many hours** humidity is high (HHR) - The **temperature during those humid hours**

Higher BU means weather is more favorable for late blight development.

FU (Fungicide Units)

A daily score representing how fast fungicide protection is “used up” or needs renewal, driven mainly by: - **Rainfall amount** - **Days since last application**

More rain and more time since spraying generally increase FU.

ABU and AFU (Accumulated BU/FU)

The model accumulates BU and FU day after day until a threshold is reached.

APP (Application count)

When thresholds indicate a spray should occur, the model increments the application counter and resets accumulations according to the SIMCAST decision rules.

Variety resistance matters

The model supports multiple “variety types” (susceptibility/resistance classes). More susceptible varieties reach thresholds sooner, typically resulting in more applications under the same weather.

Step E — Trend analysis (optional): “Is risk increasing over years?”

If you run multiple years, the pipeline can fit a **linear trend** at each location:

It models (for each point): - APP ~ year (trend in maximum number of applications) - ABU ~ year (trend in total blight pressure) - AFU ~ year (trend in total fungicide pressure)

This is done using **ordinary least squares (OLS) regression**, producing: - **Slope** (increase/decrease per year) - **p-value** (statistical significance) - **R²** (how well the linear trend explains variation)

The pipeline can then map: - Significant increase - Significant decrease - Not significant

What is the expected result if you use this?

If you run the full workflow correctly, you should expect:

- 1) **Clean, aligned climate inputs** on a consistent grid.
- 2) **Extracted time series** at each location (fast to re-use).
- 3) **Per-location SIMCAST outputs** describing:
 - Daily disease-favorable conditions (BU)
 - Daily fungicide pressure (FU)
 - Accumulated values (ABU/AFU)
 - Expected spray/application count (APP)
- 4) **Map-ready outputs** (GeoTIFF rasters) that you can visualize in GIS software.
- 5) Optionally, **trend maps** showing where risk metrics are increasing/decreasing over time.
- 6) Optionally, an **animation** summarizing patterns across “all lands” for communication.

Interpretation-wise: - Areas with higher APP typically represent **higher management pressure** (more sprays expected). - Areas with higher BU/ABU represent **more favorable disease conditions** (weather-driven risk). - FU/AFU indicate **how quickly protection may need renewal**, influenced by rainfall and time since spraying.

What are the outputs?

This project produces outputs at multiple stages. You may not need all of them depending on your goal.

1) Resampled climate files (NetCDF)

Purpose: aligned precipitation data on the target grid.

Typical filename pattern: - prec_daily_YYYY_MM.nc

What it contains: daily precipitation values on the chosen grid, for that month.

2) Extracted climate tables (Parquet)

Purpose: fast per-location climate time series used by SIMCAST.

Expected directory pattern: - <PARQUET_BASE_DIR>/<variable>/Outputs/<variable>_daily_YYYY_MM.p

Typical variables: precipitation, tmin, tmax, dew point.

Core columns (conceptually): - ID (location identifier) - FECHA (date) - Value (the climate value for that variable)

3) SIMCAST results tables (Parquet)

Purpose: the main model outputs in table form (easy to aggregate, chart, and join to geometry).

Typical output: - simcast_results.parquet per year/crop/region grouping (depending on how runs are configured)

What it contains (conceptually): For each location and day in the run period, values such as: - hr_ge_thresh: hours with RH threshold (e.g., 90%) - tmean_ge_thresh: mean temperature during those high-humidity hours - rain_mm: rainfall - tavg_C: daily average temperature - BU, FU: daily scores - ABU, AFU: accumulated scores at application decision points - APP: cumulative application count through the season

This is the most information-rich output and is the best source for further custom reporting.

4) Raster (map) outputs (GeoTIFF)

Purpose: GIS-ready surfaces to visualize patterns spatially.

Depending on which steps you run, typical GeoTIFFs include:

- **APP rasters** (application intensity / count)
 - Example pattern: `simcast_<year>_<period>_<region>.tif`
- **BU and FU summary rasters** (summed over season)
 - `BU_sum_<year>_<period>_<region>.tif`
 - `FU_sum_<year>_<period>_<region>.tif`

These rasters let you: - Make choropleth-style maps - Clip to administrative boundaries - Compare years/seasons visually

5) Trend statistics (Parquet) and trend maps (PNG)

Purpose: quantify and visualize long-term changes over multiple years.

Typical outputs: - A table of trend stats per location (e.g., slope/p-value/R² for APP/ABU/AFU) - PNG maps showing: - significant increase/decrease/not-significant - coefficient (slope magnitude) for significant points

6) Animation outputs (visual summary)

Purpose: communication product for stakeholders, showing changes across space over time.

The pipeline can assemble yearly/seasonal rasters into an “all lands” animation.

How should a non-coder user read the results?

A practical guide:

- If you want a **single seasonal risk map**, use the **GeoTIFF outputs** (APP, BU_sum, FU_sum).
- If you want **numbers per location** (for tables, dashboards, reports), use the **Parquet outputs**.
- If you want to answer “Is risk increasing?”, use the **trend outputs** (trend parquet + PNG maps).

A good validation approach (recommended): - Pick a few locations and plot their daily outputs (BU/FU/APP) to check the seasonality looks plausible. - Compare spatial patterns against known agroecological gradients (elevation, wet/dry zones). - If available, compare with historical disease observations or spray records.

Summary (answers to your required questions)

What is it?

A spatial SIMCAST pipeline that converts gridded climate data into late blight risk and fungicide-application indicators across many locations.

What is it for?

To support potato late blight risk mapping, agroclimatic zoning, seasonal decision support, and multi-year trend analysis.

What is the expected result if you use this?

You will obtain per-location and map-ready outputs (tables + rasters) describing humidity-driven disease favorability (BU/ABU) and expected application pressure (APP), optionally with trend statistics over years.

What are the outputs?

NetCDF resampled precipitation, Parquet climate extracts, Parquet SIMCAST results, GeoTIFF rasters (APP/BU/FU summaries), optional trend Parquet + PNG maps, and optional animations.